Yanze Li

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RESEARCH INTERESTS

I'm a Ph.D. student at University of British Columbia (since Fall 2021), advised by Alexander J. Summers and Ivan Beschastnikh. My research interests lie in programming languages (especially language semantics, type systems, and type theory) and formal verification (program logics, theorem proving, and deductive verification). I'm particularly interested in how we can write simple and intuitive specifications about various properties of programs and formally prove their correctness.

EDUCATION

Ph.D.	Computer Science, University of British Columbia, 2021 - Now *Advisor: Alexander J. Summers, Ivan Beschastnikh*
M.S.	Computer Science, Texas A&M University, 2020 *Thesis: Efficient and Scalable Whole Program Race Detection for Java and Android Programs* *Advisor: Jeff Huang*
B.Eng.	Electrical Engineering, Huazhong University of Science and Technology, 2017

PUBLICATIONS

ICSE'22	*"PUS: A Fast and Highly Efficient Solver for Inclusion-based Pointer Analysis"* Peiming Liu, **Yanze Li**, Bradley Swain, Jeff Huang *International Conference on Software Engineering (ICSE'22). 2022.* ***ACM SIGSOFT Distinguished Paper Award***
Correctness'21	*"OpenRace: An Open Source Framework for Statically Detecting Data Races"* Bradley Swain, Jeff Huang, Bozhen Liu, Peiming Liu, **Yanze Li**, Addison Crump, Rohan Khera *2021 IEEE/ACM 5th International Workshop on Software Correctness for HPC Applications (Correctness). IEEE, 2021.*
PLDI'21	*"When Threads Meet Events: Efficient and Precise Static Race Detection with Origins"* Bozhen Liu, Peiming Liu, **Yanze Li**, Chia-Che Tsai, Dilma Da Silva, Jeff Huang *42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation. 2021.*
SC'20	*"OMPRacer: A Scalable and Precise Static Race Detector for OpenMP Programs"* Bradley Swain, **Yanze Li**, Peiming Liu, Ignacio Laguna, Giorgis Georgakoudis, Jeff Huang

	International Conference for High Performance Computing, Networking, Storage and Analysis. IEEE, 2020.
ICSE'19	(Demo Track) *"SWORD: A Scalable Whole Program Race Detector for Java"* **Yanze Li**, Bozhen Liu, Jeff Huang *2019 IEEE/ACM 41st International Conference on Software Engineering: Companion Proceedings (ICSE-Companion). IEEE, 2019.*

RESEARCH EXPERIENCE

2021.9-Now **Research Assistant, University of British Columbia, Canada**
Worked on automated verification of Rust async runtimes, especially their responsiveness (a kind of liveness) properties.
I'm currently working on the formal verification of eBPF and its verifier. I used F* to formally model the semantics of eBPF bytecode and the Linux kernel verifier with the goal of developing sound and efficient eBPF verifiers.

2020.8-2021.6 **Research Intern (Remote), Utrecht University, Netherland**
Worked with Dr. Jurriaan Hage on the LLVM backend and FFI of a Haskell compiler called Helium.

2018.6-2020.6 **Research Assistant, Texas A&M University, USA**
Worked on static analysis for concurrent programs. Developed tools that scale to million lines of Java/C++/Android code and efficiently detect potential data races and deadlocks.

WORK EXPERIENCE

2024.9-2024.12 **Applied Scientist Intern, AWS, USA**
I worked on a randomized concurrency testing tool called Shuttle. I developed a lightweight library for annotating large codebases to measure customized state coverage when running Shuttle. I also worked on applying Reinforcement Learning to improve state coverage when doing concurrency testing.

2019.7-2021.5 **Software Engineer, Sec3 (previously Coderrect), USA**
(As intern during 2019.7-8, 2020.2-7, as full-time employee during 2020.10-2021.5)
Worked as the main developer of an LLVM-based program analysis tool for detecting concurrency bugs and anti-patterns in C/C++/Fortran/CUDA code. I designed a highly efficient static happens-before graph, lock tracking and race detection algorithms which enable the tool to analyze million lines of code in minutes with low false positive rate.

2015.11-2017.4 **Software Engineer, Nightingale Technology, China**
Worked on a second-hand commodities trading platform for college students and an integrated web application for editing and publishing news articles as well as managing and visualizing their statistics.

TEACHING EXPERIENCE

2024S CPSC 410: Advanced Software Engineering, Teaching Assistant

2023F	CPSC 539S: Program Verifiers and Program Verification, Teaching Assistant
2022F	CPSC 410: Advanced Software Engineering, Teaching Assistant
2022S	CPSC 416: Distributed Systems, Teaching Assistant
2021F	CPSC 410: Advanced Software Engineering, Teaching Assistant

PROJECTS

Below is the list of projects I've worked on:

Shuttle A library for (randomized) concurrency testing for Rust code. [GitHub]

TraceChecker A DSL for distributed system trace verification. Used as a teaching tool to grade students' assignments based on formal specifications. [GitHub]

LTLSpec A proof-of-concept Haskell framework for modelling, specifying, and verifying distributed system traces in linear temporal logic. [GitHub]

Helium A compiler for a subset of Haskell that aims at delivering high quality type error messages particularly for beginner programmers. It also includes facilities for specializing type error diagnosis for embedded domain specific languages. [GitHub]

Coderrect An LLVM-based static analyzer, specialize in detecting concurrency related bugs and anti-patterns, found several previously unkown bugs in Linux kernel, Redis, memcached, and GraphBLAS. [Website] [GitHub]

OMPRacer An LLVM-based race detector for OpenMP programs, using inter-procedure value-flow analysis to reason about array accesses. Found several previously unknown bugs in ECP proxy applications and a major simulator for COVID-19. [GitHub]

Crappie An incremental race detection engine that scales to distributed systems and Android apps and has been implemented as an Intellij IDEA plugin.

SWORD A whole program race detector for Java (source code/bytecode) and has been implemented as an Eclipse plugin. [GitHub]

HONOR AND AWARDS

2022	ACM SIGSOFT Distinguished Paper Award
2022	OPLSS Fellowship Grant
2019	ACM SIGSOFT CAPS Award
2017	Excellent Graduated Student at HUST
2015	Scientific Research Innovation Scholarship
2014	3rd place, China University Cloud Computing Innovation Competition

SERVICE

2020.8-2022.11	SIGPLAN Long-Term Mentoring Program (SIGPLAN-M), Operations Team

SUB-REVIEWER

2027	PLDI
2026	ICSE
2023	ICSE
2022	ASE
2020	OOPSLA
2019	PLDI, ICSE, FSE, OOPSLA
2018	TSE